

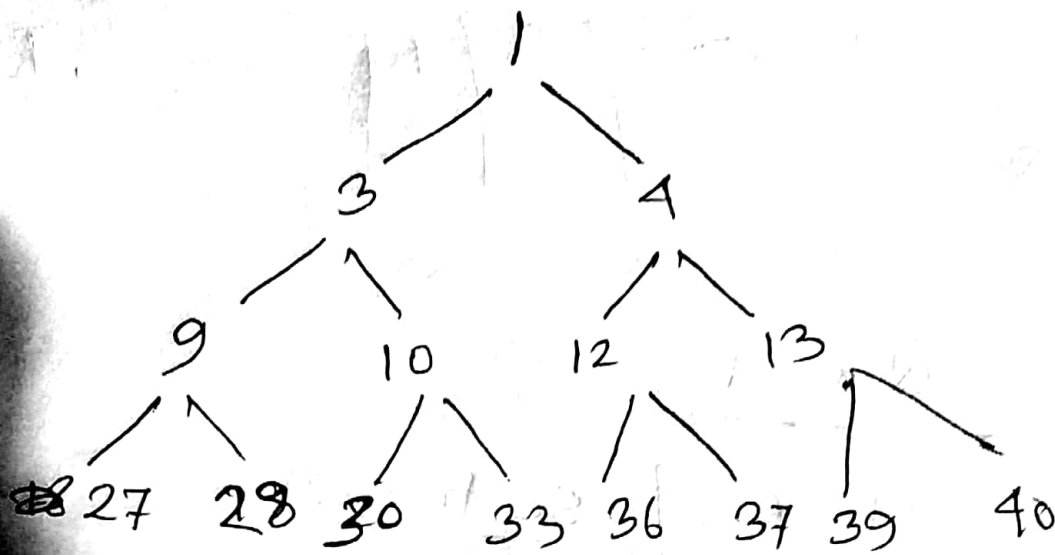
EX 1

Quiz-1

Set - A

Section - A

- a) False. A Lucky DFS might expand exactly nodes to reach the goal. A* largely dominates graph-search algorithm that is guaranteed to find optimal solutions.
- ¹ a) True. $h(n) = 0$ is always an admissible heuristic since costs are nonnegative.
- b) False. A* search is often used in robotics, space can be discretized or skeletonized.
- ~~a)~~ c) True. Depth of the solution matters for b first search, not cost.
- d) False. A rook can move ~~across across~~ across the board in move one, although the Manhattan distance from start to finish is 8.



c) ~~of~~ Bidirectional search is very useful, because the only successor of n in the reverse direction is $\lfloor (n/2) \rfloor$. This helps focus the search. The branching factor is 2 in the forward direction; 1 in the reverse direction.

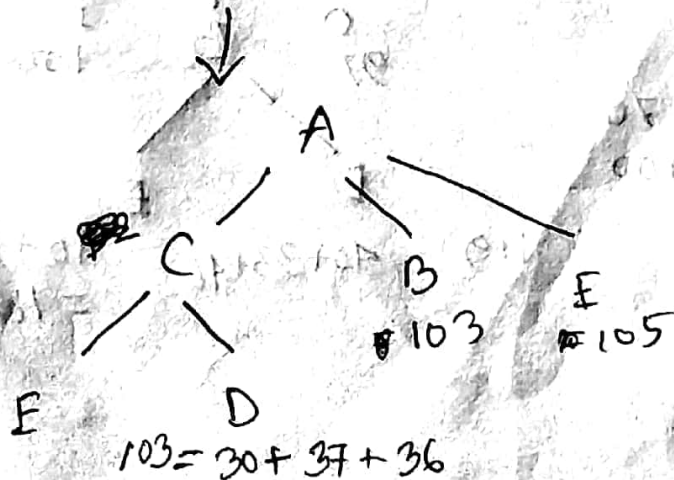
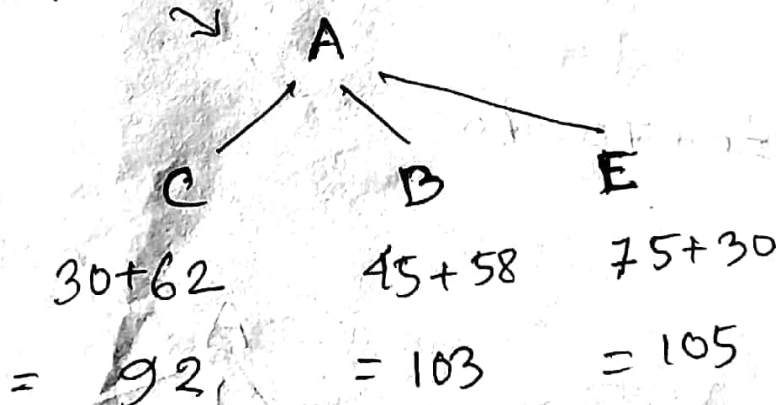
1. IDS combines the small memory footprint of DFS, and has the completeness guarantee of BFS.

In BFS space complexity is $O(b^d)$, where b is branching factor and d is depth. In IDS space complexity is $O(bd)$. So, the memory is smaller for IDS.

DFS may not be complete if it has infinitely deep paths with no solution. IDS solves this problem as it is complete like BFS.

2. $A = 0 + 65 = 65$

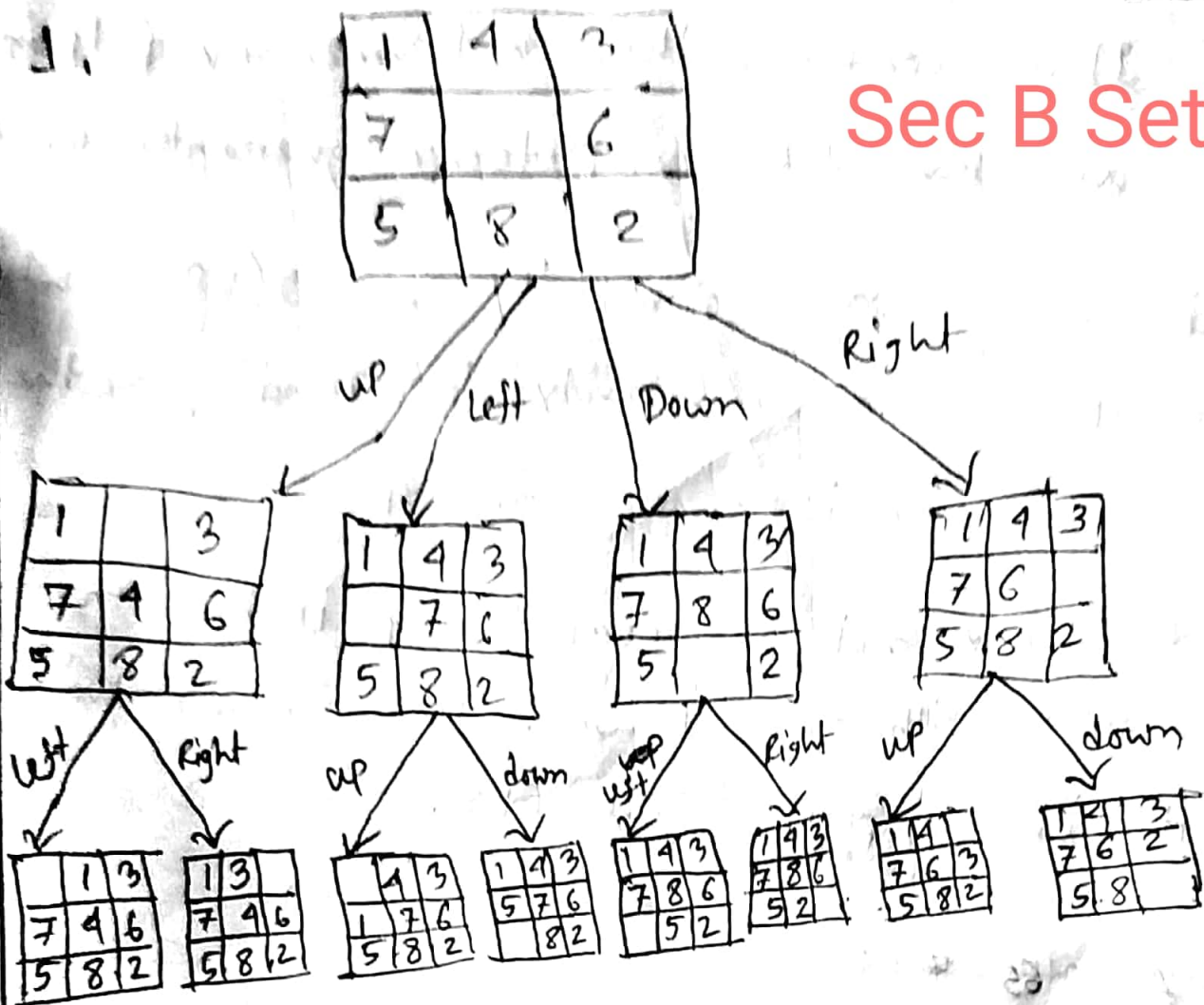
Sec B Set A



$96 = 30 + 36 + 30$

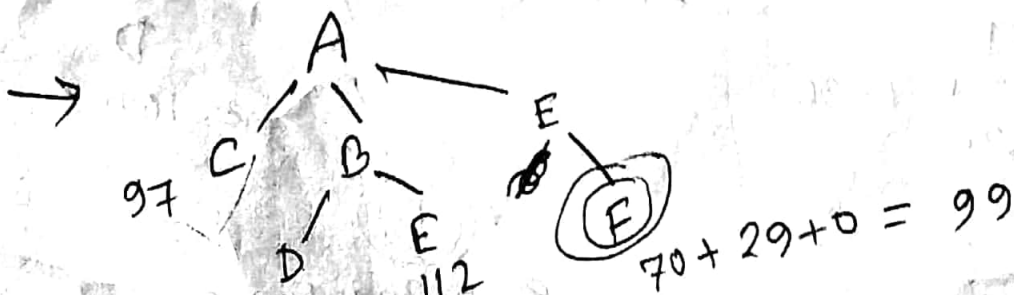
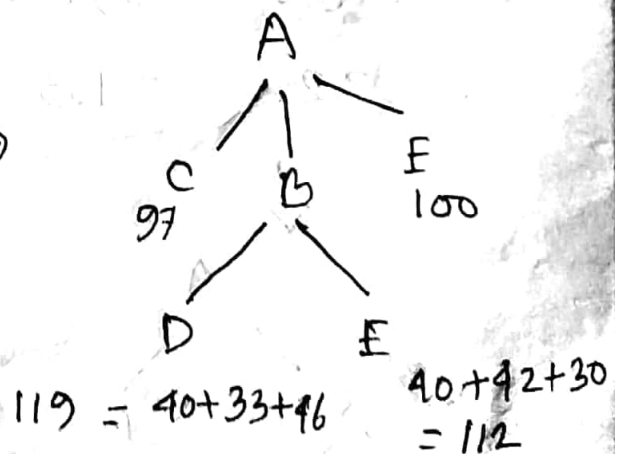
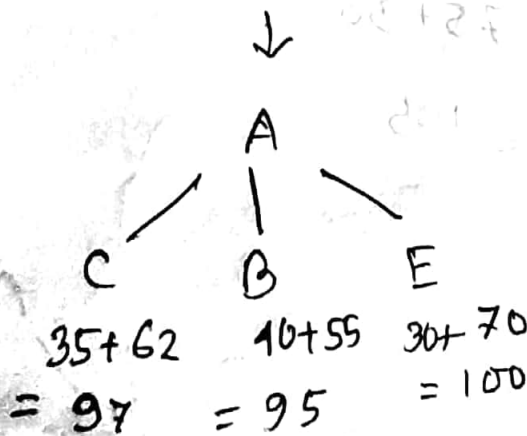
$96 = 30 + 30 + 36$

Sec B Set B



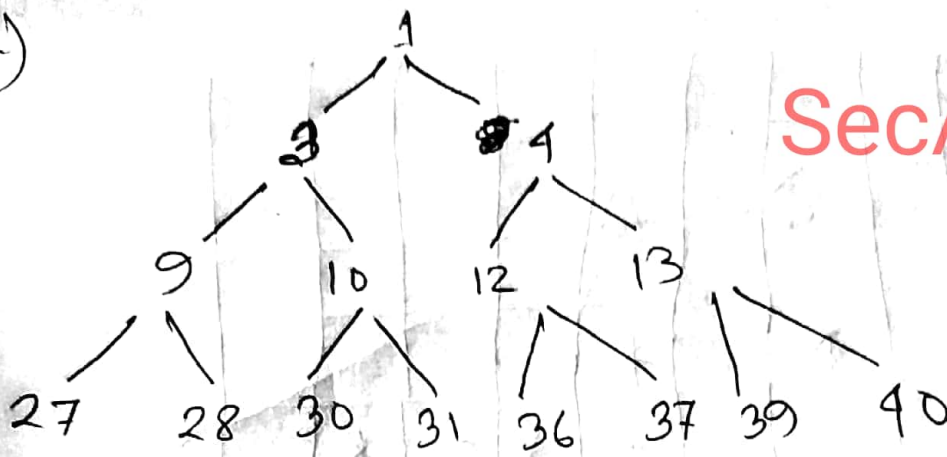
2.

$$A = 0 + 70 = 70$$



SecA Set A

2 a)



b)

BFS - 1, 3, 4, 9, 10, 12, 13, 27, 28, 30, 31, 36, 37, 39, 40.
no goal state

Dfs with limit 3

1, 3, 9, 27, 28, 10, 30, 31, 4, 12, 36, 39,

no goal state

c) Iterative deepening : ~~1, 3, 4, 9, 10, 12, 13, 27, 28, 30, 31, 36, 37, 39, 40~~

1
1 3 4

1 3 9 10 4 12 13

1 3 9 27 28 10 30 31 4 12 36

no goal state